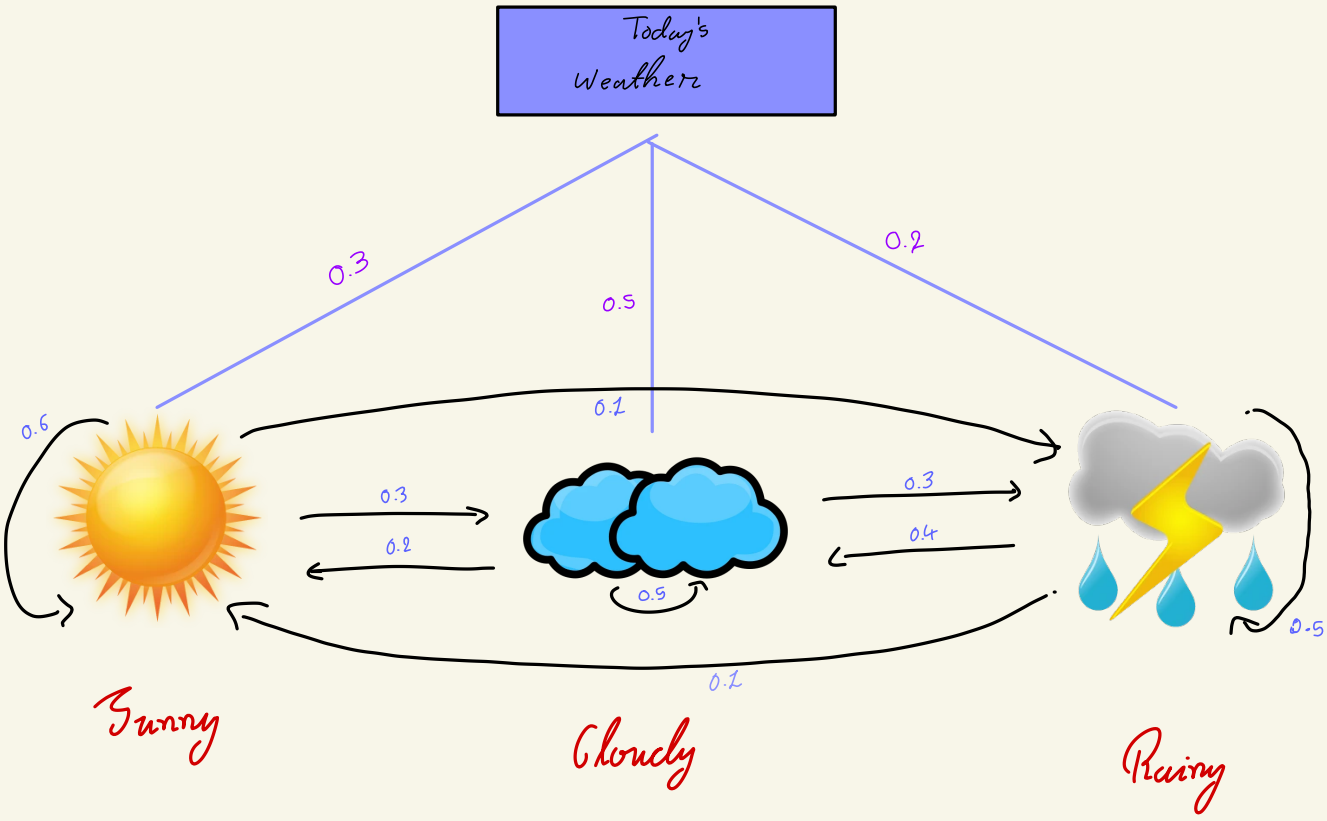
Uncertainity Probabilistic Interface (Markov Model)

- It helps to modeling uncertainity with probability.
- Helps to make prediction based on sequential data.

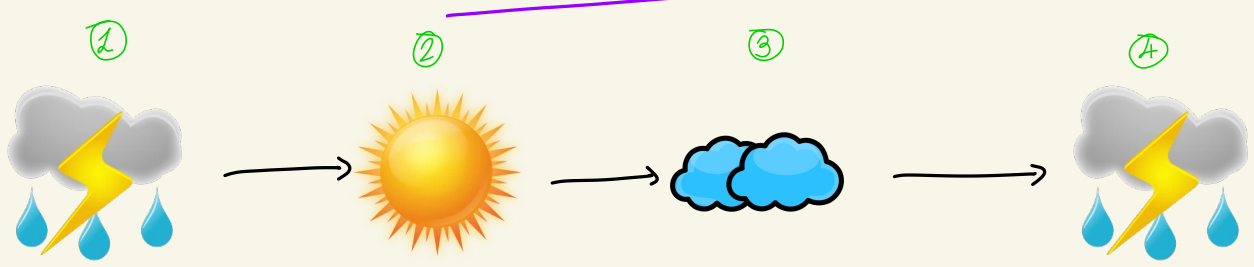


Sunny	Cloudy	Rainy
0.3	0.5	0.2

	Sunny	Cloudy	Rainy
Sunny	0.6	0.3	0.1
Cloudy	0.2	0.5	0.3
Rainy	0.1	0.4	0.5

Tomorrow is depending on present event.

Problem statement



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1. Today's weather = Rainy = 0.2

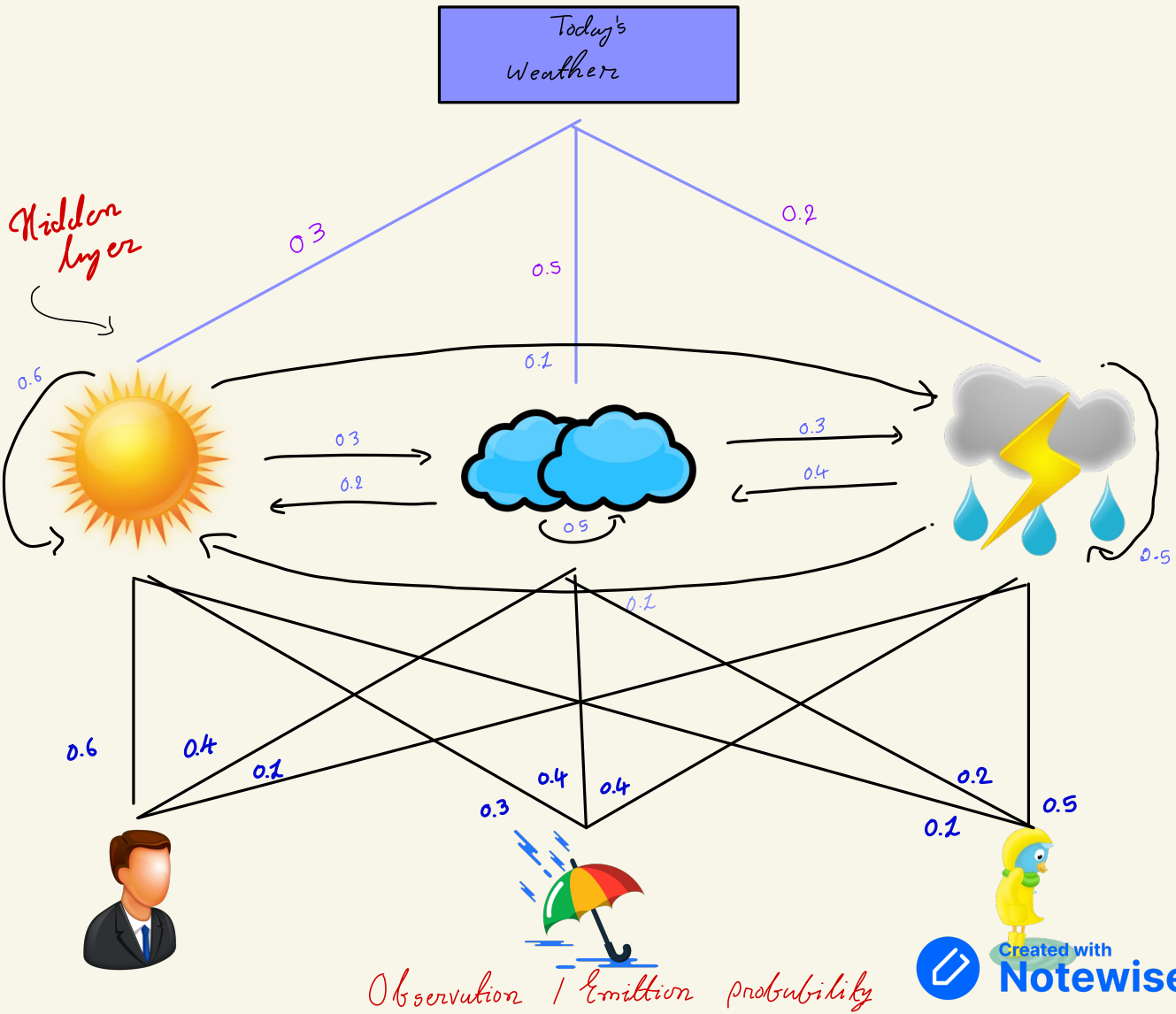
2. Tomorrow's weather = Sunny = $P(\text{Sunny}/\text{Rainy})$ = 0.1
(depends on current weather)

3. Tomorrow's weather = Cloudy = $P(\text{Cloudy}/\text{Sunny})$ = 0.3
(depends on current weather)

4. Tomorrow's weather = Rainy = $P(\text{Rainy}/\text{Cloudy})$ = 0.3
(depends on current weather)

$0.2 \times 0.1 \times 0.3 \times 0.3$

Hidden Markov Model



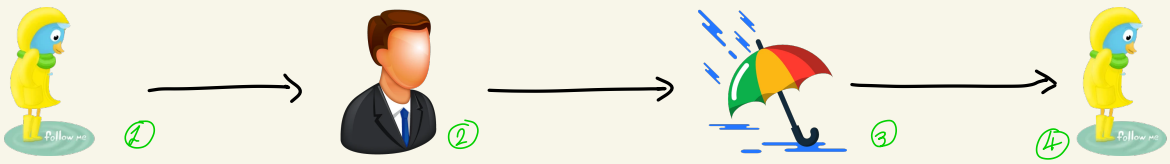
Created with Notewise

Sunny	Cloudy	Rainy
0.3	0.5	0.2

	Sunny	Cloudy	Rainy
Sunny	0.6	0.3	0.1
Cloudy	0.2	0.5	0.3
Rainy	0.1	0.4	0.5

	Normal	Umbrella	Raincoat	
Sunny	0.6	0.3	0.1	= 1
Cloudy	0.4	0.4	0.2	= 1
Rainy	0.1	0.4	0.5	= 1

Problem statement



Raincoat Normal Umbrella Raincoat

1. Person came out with raincoat, the probability of weather would be either Sunny or Rainy or Cloudy.
2. Person came out as normally, the probability of weather would be either sunny, cloudy or rainy.
3. Person came out with umbrella, the probability of weather would be either sunny, cloudy or rainy.
4. Person came out with raincoat, the probability of weather would be either sunny, cloudy or rainy.

by combining all results, we have total 81 cases.
$(3 \times 3 \times 3 \times 3)$

Possible combinations

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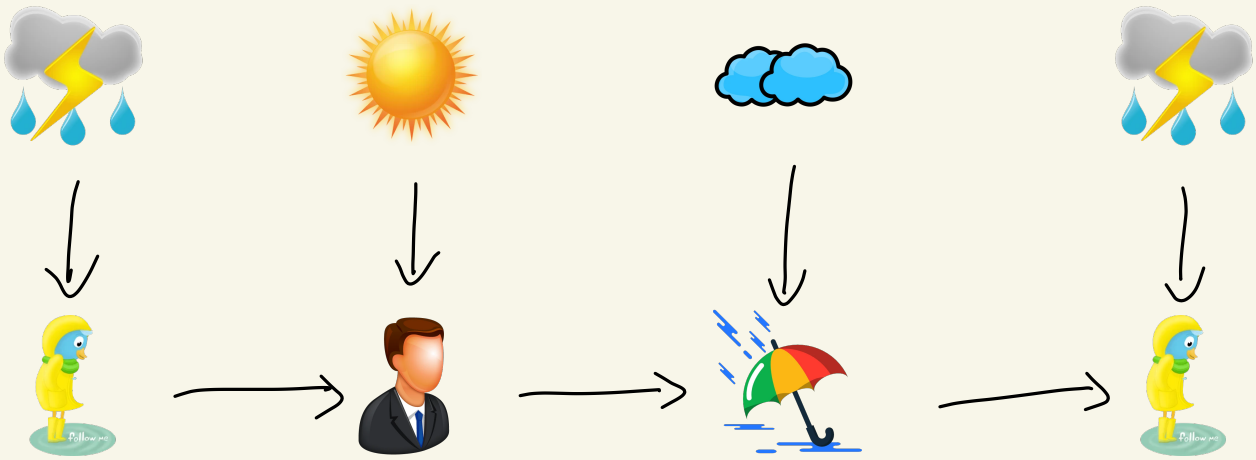
C → R → C → S

R → C → R → S

⋮

- At the end, we will get total 81 unique combinations.
- The answer will be that one combination, whose probability is highest.

Single sample combination



weather = rainy & Person in raincoat

weather = sunny & Person is in normal state

weather = cloudy & Person with umbrella

weather = rainy & Person in raincoat

R → S → C → R

$P(\text{Raincoat} / \text{Rainy}) \times P(\text{Normal} / \text{Sunny}) \times P(\text{Umbrella} / \text{Cloudy}) \times P(\text{Raincoat} / \text{Rainy})$

$(0.5 \times 0.6 \times 0.4 \times 0.5)$

$\times$

$($ 0.2 $\times$ 0.1 $\times$ 0.3 $\times$ 0.3

Today is rainy

$P(\text{Sunny} / \text{Rainy})$

$P(\text{Cloudy} / \text{Sunny})$

$P(\text{Rainy} / \text{Cloudy})$

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0.06 × 0.0018 = 0.000108

Now let's see the role of Hidden Markov Model in POS (Parts-of-speech) tagging.

Training data

1. Parth loves CodeWithHarry
2. Can parth google CodeWithHarry
3. Will riya google CodeWithHarry
4. riya loves will.

Steps

1. Take training data
2. Apply POS tagging on training data
3. Train Hidden Markov Model on training data
4. Do pos tagging by Hidden Markov-Model on test sentence.

POS tagging

1. Parth (Noun) loves (Verb) CodeWithHarry (Noun)
2. Can (model) parth (Noun) google (verb) CodeWithHarry (Noun)
3. Will (model) riya (Noun) google (verb) CodeWithHarry (Noun)
4. riya (Noun) loves (verb) will. (Noun)
5. will (Noun) loves (verb) google (Noun)

Emittion probability →

	Noun	Verb	Model
parth	2	–	–
loves	–	3	–
CodeWithHarry	3	–	–
can	–	–	1
google	1	2	–
will	2	–	1
riya	2	–	–

converting into probabilities.

Exc. if a word is noun then, what is the probability that the word is Parth.
if a word is verb then, what is the probability that the word is Parth.
if a word is model then, what is the probability that the word is Parth
⋮
if a word is model then, what is the probability that the word is riya.

The probability of Noun is parth = 2/10

The probability of Noun is CodeWithHarry = 3/10

Now we have Emission probability..!

	Noun	Verb	Model
Parth	2/10	–	–
loves	–	3/5	–
codewithharry	3/10	–	–
can	–	–	1/2
google	1/10	2/5	–
will	2/10	–	1/2
riya	2/10	–	–

Transition probability ↓

Created with Notewise

- Based on training data
- What is the probability of next word will be Verb if current word is Noun.
- What is the probability of next word will be Noun if current word is Verb.

Steps

1. Add 'S' (start) prefix on starting of sentence. & 'E' (end) postfix on end of sentence.

S Noun Parth Verb loves Noun CodeWithHarry E

S Model Can Noun parth Verb google Noun CodeWithHarry E

S Model Will Noun raja Verb google Noun CodeWithHarry E

S Noun raja Verb loves Noun will. E

S Noun will Verb loves Noun google E

2. Maintain a look-up table.

	Noun	Verb	Model	E
S	3	—	2	—
Noun	—	5	—	5
Verb	5	—	—	—
Model	2	—	—	—

- Now count the frequency such as....
  - "How many times Noun came immediate after S."
  - "How many times Verb came immediate after S."
  - "How many times Model came immediate after S."

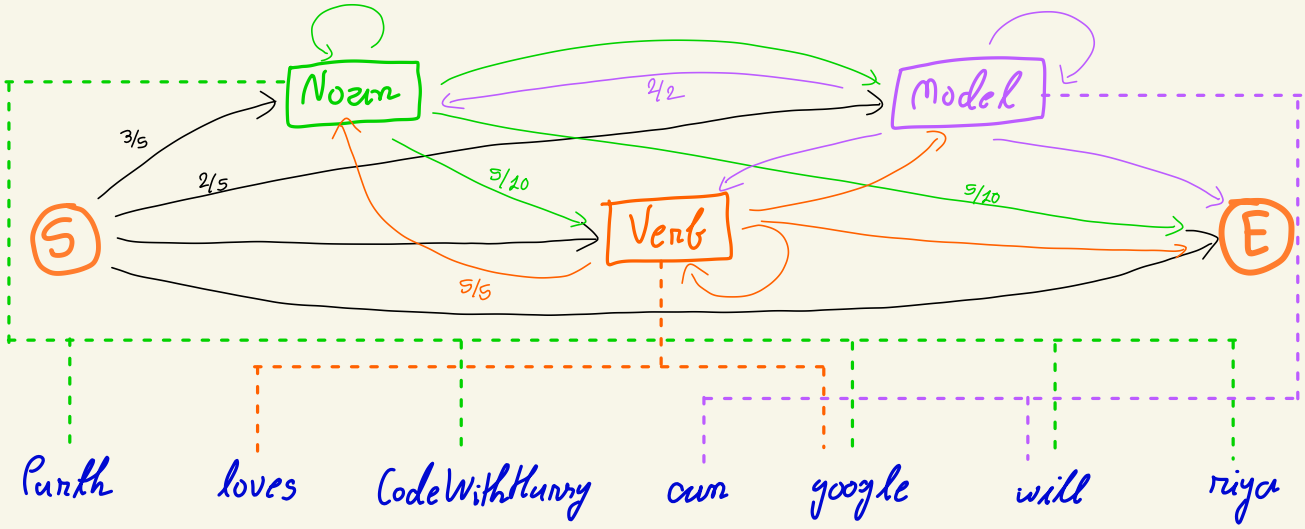
converting into probabilities

	Noun	Verb	Model	E
S	3/5	—	2/5	—
Noun	—	5/10	—	5/10
Verb	5/5	—	—	—
Model	2/2	—	—	—

Now we calculated transition probabilities...!

Now Let's design our hidden markov model using Emission & transition probabilities.

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Test sentence : will will google CodeWithHarry.

Now we will try every possible combinations. $pos^{words} = 3^4 = 81$

combination 1

will will google CodeWithHarry.

Emission Probability

Transition Probability

S → N → N → N → N → E

3/5 | 2/10 | 2/10 | 1/10 | 3/10 | 5/10

0 | 0 | 0

$(2/10 \times 2/10 \times 1/20 \times 3/10) \times (3/5 \times 0 \times 0 \times 0 \times 5/10) = 0$

combination 2

will will google CodeWithHarry.

S → N → N → N → M → E

3/5 | 2/10 | 2/10 | 1/10 | 0 | 0

0 | 0 | 0

$(2/10 \times 2/10 \times 1/10 \times 0) \times (3/60 \times 0 \times 0 \times 0 \times 0) = 0$

combination 3

will will google CodeWithHarry.

S → M → N → V → N → E

2/5 | 1/2 | 2/10 | 2/5 | 3/10 | 5/10

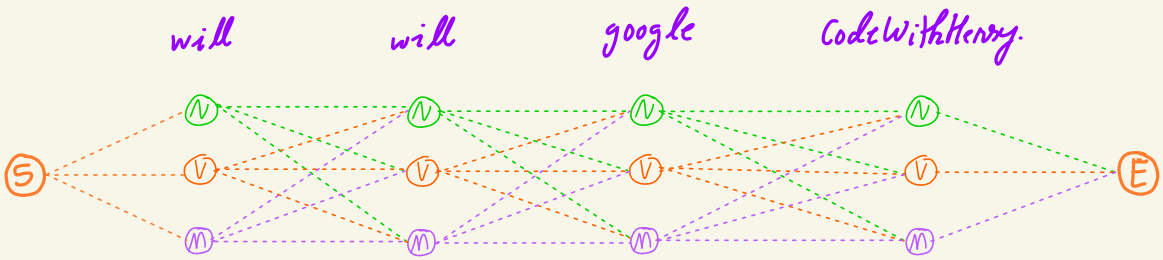
2/2 | 5/10 | 5/5 | 5/20

$(1/2 \times 2/10 \times 2/5 \times 3/20) \times (2/5 \times 2/2 \times 5/10 \times 5/5 \times 5/20)$

$\frac{12}{2000} \times \frac{500}{5000}$

$3/250 \times 1/10 = \boxed{3/2500} = \boxed{0.0012}$

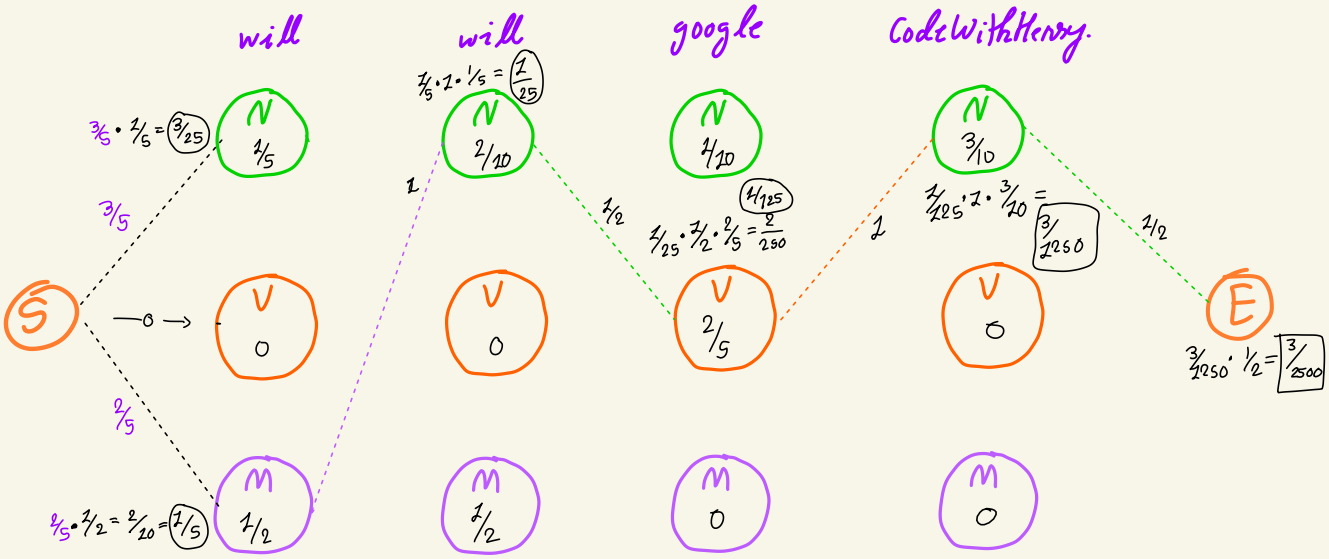
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$3^4 = 81$ combinations

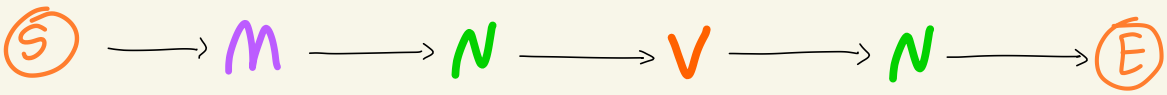
- This number will rise exponentially as sentence length gets bigger
- Hence it uses an optimization technique/algorithm named as Viterbi Algo.

Viterbi Algorithm



- Start with (S) & check transition probability of all possibilities.

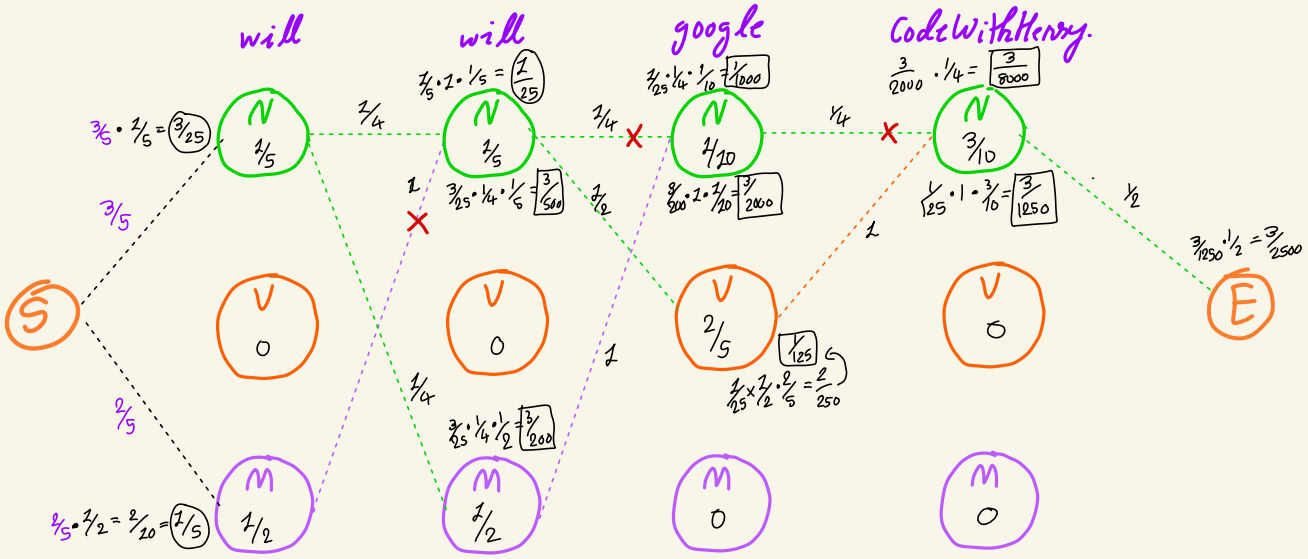
final path



for experiment purpose:

	Noun	Verb	Model	E
S	3/5	—	2/5	—
Noun	1/4	5/10	1/4	5/10
Verb	5/5	—	—	—
Model	2/2	—	—	—

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• At any node, if we got 2 probability values, then we will select bigger one.

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